Advanced Compute and Data Core (ACDC): From Core Facility Data Generation to FAIR Research Objects

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Abstract

Visual omics is an emerging research frontier, where the integration of bioimaging and omics data, to study spatial and temporal processes in cells and tissues, is leading to new integrated analysis methods. The Advanced Compute and Data Core (ACDC) has been established at the Amsterdam University Medical Centre to develop a data and compute infrastructure that will support FAIR visual omics and multi-omics research. In collaboration with the core facilities in omics and bioimaging, ACDC is developing an AI mediated research infrastructure that is FAIR by design and supports the data life cycle, to produce and reuse visual omics research objects.

Keywords

FAIR research infrastructure, FAIR by Design, ISA, Research Objects

1. Introduction

Advances in microscopy and omics laboratory methods are driving the development of new integrated bioimaging and multi-omics analysis approaches for studying spatial and temporal biological processes. As these approaches require the use of multiple types of omics and bioimaging technologies, and complex, big data analysis approaches, researchers require infrastructure to facilitate FAIR data integration and analysis on high performance computational resources. The Advanced Compute and Data Core (ACDC) infrastructure, has been established to support this kind of FAIR multi-omics and visual omics research, and provides a shared data and compute environment for researchers using the Amsterdam UMC Core Facilities.

ACDC supports the whole data life cycle, from planning experiments and collecting data, through to pre-processing, analysis, archiving and sharing. The infrastructure is FAIR by Design. It assists researchers by automating data management processes from the moment of data generation at the core facilities, lowering the barriers to creating FAIR data, and increasing the opportunities to exploit the results of such integrated results. This is made possible, in part, by developing new AI methods, based on Retrieval-Augmented Generation (RAG) tools and

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language models, for structuring and automating FAIR data practices, to enable greater integration and reuse.

The ACDC infrastructure is an ecosystem of established tools and resources, which are reused and extended, to address specific requirements. It shares components with several ELIXIR RDMKit Tool Assemblies [1], building on established FAIR data management solutions, which include interconnected components for metadata and data management, scientific workflows, and high performance computational resources.

2. Methods and Results

The development of the ACDC infrastructure follows three key design principles:

1. Reuse established components – build on existing tools, resources and community standards, to accelerate progress and to benefit from community experience
2. Incremental approach – piloting specific solutions to data and compute bottlenecks experienced by one core facility, before making solutions more generally available
3. Minimising extra work – automation of routine processes and assistance from AI approaches, making it easier for individual researchers to create and use FAIR data

The **FAIRDOM-SEEK** [2] is the central **metadata management** component for ACDC, allowing Amsterdam UMC researchers to share (meta)data in the context of their whole experiments, interlinking multi-omics and bioimaging, across different types of research assets. The standardisation and semantic integration of metadata is essential to supporting the whole data life cycle. Multiple different types of metadata must be brought together, including the experimental design, biological samples, laboratory instruments, pre-processing and processing of data, modelling, analysis and provenance.

The FAIRDOM SEEK provides semantic integration via the underlying *Just Enough Results Model* (JERM) ontology, which describes research assets and the connections between them. The JERM encapsulates the core components of the Investigations, Studies and Assay (ISA) standard, for interlinking different omics datasets. Tabular templates, conforming to the JERM and ISA, and mapping to domain-specific metadata models (e.g. REMBI for bioimaging and MIAPE for proteomics), enable researchers to more easily use the most appropriate community standards. The use of the JERM facilitates automated metadata extraction into RDF (Resource Description Framework) when research assets are registered in the FAIRDOM SEEK, for richer semantic querying.

The FAIRDOM SEEK is integrated with the **FAIR Data Station** (FDS) [3], to facilitate the generation of FAIR metadata templates. The FDS is a metadata ingestion platform that can define and validate metadata schemas, providing researchers with easy to configure and use tabular templates that will produce valid ISA metadata.

The FAIRDOM-SEEK manages metadata in context, wherever the underlying data may reside. Data storage is managed with the Integrated Rule-Oriented Data System (iRODS https://irods.org/), which can assist with tiering and data migration throughout the data life cycle. It enables secure collaboration and virtualizes data storage resources. iRODS is the

intermediate layer between the FAIRDOM-SEEK and local & remote high performance compute infrastructures, where advanced users can interact directly with their data during analysis workflows. A graphical user interface is provided by ManGO to simplify management of the data analysis processes.

The Amsterdam UMC developed a local High Performance Compute (HPC) resource, to provide a compliant and secure infrastructure for highly parallel, large memory and GPU workloads. The HPC is available to all Amsterdam UMC employees and is supported by a team of experts to facilitate the development of efficient and reproducible workflows for clinical research and diagnostic applications.

Here we show how the HPC, iRODS, FAIR Data Station and FAIRDOM-SEEK interact to enable sharing and reuse to support and enable FAIR visual omics experiments. Experimental metadata is shared in the FAIRDOM-SEEK, according to ISA metadata templates created and validated using the FAIR Data Station. Data files generated during the experiments are managed by iRODS, with technical metadata relating to the files and their properties made available to researchers via the FAIRDOM SEEK. File locations in iRODS are also available via the HPC, so researchers can run their analyses and integrate their results with the ISA metadata descriptions. The result is increased sharing of research assets, fostering collaboration and enabling the generation and reuse of FAIR research objects.

Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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